AYESHA RAHMAN

AI / Machine Learning Engineer

DETAILS

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SUMMARY

AI/ML Engineer with an MEng in Computer Science with Artificial Intelligence, specializing in secure and scalable AI systems across cybersecurity, cloud infrastructure, fintech and data-driven applications. Experienced in building production-grade machine learning pipelines, privacy-preserving and federated learning solutions, anomaly detection systems, secure APIs and cloud-based platforms using Python, PyTorch, TensorFlow, AWS, Azure, GCP, Docker, FastAPI, Flask and React. Strong focus on Zero Trust architecture, explainable AI, compliance-ready data handling and turning research-driven AI solutions into reliable real-world systems.

EDUCATION

MEng, University of Leeds, United Kingdom
2024 - 2025
Strong 2:1 overall, with 81% in dissertation.

BSc, University of Leeds, United Kingdom
2021 - 2024
Second-Upper Class Honors (2:1).

EXPERIENCE

Research Intern and Mentorship, CyberSaR (KAUST), Remote, Jeddah, Saudi Arabia
Dec 2025 - Present

- Developed behavioural authentication using human and process-level telemetry with VAE, Transformer and CNN models.
- Built scalable pipelines and TSFresh feature extraction across CERT, LANL and IAM datasets.
- Applied federated learning, differential privacy, agentic AI and XAI for interpretable risk scoring.
- Developed MVP and POC for system development.

Freelance Software Developer, Outlier, Leeds, United Kingdom
Dec 2024 - Sep 2025

- Designed and tested AI prompts with focus on edge cases and QA.
- Developed full-stack websites using HTML, CSS, JavaScript and Flask.
- Managed cross-border teams to deliver client web tools.

Equity Research IT Analyst, UTI Mutual Fund, Mumbai, India
Jul 2023 - Sep 2023

- Engineered two Python web scraping tools for financial market analysis and secure data handling.
- Delivered Travel/Hospitality sector insights supporting investment decisions.
- Reduced manual data processing effort by 30% through Python analytical scripts.

SKILLS

- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, Time-Series Forecasting, Anomaly Detection, Federated Learning, Explainable AI, RAG, Agentic AI, Multimodal AI
- **Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, LangChain, FAISS, ONNX, OpenVINO, TensorFlow Lite
- **Models:** CNN, RNN, LSTM, GRU, Transformer, GNN, Autoencoders, SARIMAX, ARDL, N-BEATS, N-BEATSx,

PROJECTS

AgenCAT: Agentic Continuous Adaptive Trust for Post-Login Identity Assurance *(ongoing)*
Designed a Zero Trust framework of local identity agents that maintain per-user digital twins to continuously re-evaluate session trust after login, closing the post-login gap where valid credentials keep operating as compromise emerges. Implemented Welford online behavioural twins with global-local score blending, cold-start warmup, a temporal convolutional-attention autoencoder for account-level trust trajectories, and a policy engine driving step-up MFA, session restriction and blocking.

Federated Behavioural Anomaly Detection for Identity & Access Control
Built a privacy-preserving detection pipeline over authentication-path and process-level telemetry, mapping raw logs into progressive feature spaces and scoring them with an ensemble of autoencoder, Isolation Forest, LightGBM and logistic-regression models. Applied federated learning (FedAvg), differential privacy (DP-SGD), explainable AI and tamper-evident audit logging, benchmarking against real red-team labels on the LANL Cyber1 and RBA corpora.

A Comparison of Inflation Forecasting Models *(MEng Diss)*
Developed multi-horizon U.S. inflation forecasting models using PCEPI and macroeconomic indicators across statistical, machine learning and deep learning approaches. Implemented ARDL, N-BEATS, N-BEATSx and N-HiTS with walk-forward validation, Fourier features, lag features, rolling statistics, scaling and exogenous variable handling. N-HiTS achieved the strongest performance across horizons, outperforming statistical, ML and deep learning baselines.

Machine-Learning-Driven Loop Optimization Selector for Compiler Auto-Tuning
Developed an ML-based compiler optimization module to automatically select loop transformations including tiling, unrolling, vectorization and OpenMP. Built a full pipeline for data collection, feature extraction, model training, benchmarking, SHAP explainability and deployable optimization schedules.

Automated Literature Review & LLM Efficiency Evaluation Platform
Designed a modular research assistant for literature mining, retrieval-augmented generation, summarisation and CPU-only LLM efficiency benchmarking. Implemented MiniLM, FAISS/NumPy, hybrid dense and TF-IDF reranking, FLAN-T5, LoRA/quantisation experiments, Streamlit dashboards and reproducible reporting.

Smart Poultry Farm Management Desktop Platform
Engineered a secure desktop platform for broiler farm management with encrypted storage, RBAC, credential rotation, audit trails, KPI reporting and automated PDF/CSV exports. Built a multithreaded simulator for environmental telemetry with real-time analytics, anomaly detection, mortality-risk scoring and dashboard visualization.

N-HiTS, NeRF

- **Cybersecurity:** Zero Trust, IAM, RBAC, OAuth2, SSO, JWT, OWASP, HTTPS/TLS, Encryption, SQLCipher, Argon2id/Bcrypt, Audit Trails, Threat Modelling, GDPR-aware data handling
- **Cloud & DevOps:** AWS, Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, EC2, S3, Lambda, CloudFront
- **Programming & Web:** Python, C++, C, Java, JavaScript/TypeScript, SQL, Bash, FastAPI, Flask, React, Streamlit, REST APIs, SQLAlchemy
- **Data & Analytics:** PostgreSQL, MySQL, MongoDB, Redis, Pandas, NumPy, Matplotlib, Plotly, Feature Engineering, Statistical Modelling
- **Edge, Robotics & Vision:** ROS2, OpenCV, LiDAR-camera fusion, SLAM, Gazebo, NVIDIA Jetson, Intel Movidius, Edge Deployment

CERTIFICATIONS AND LICENSES

- Microsoft Certified: Security, Compliance and Identity Fundamentals
- Microsoft Certified: Azure AI Fundamentals
- Microsoft Certified: Azure Fundamentals
- Blockchain Transformations of Financial Services (INSEAD)
- Python for Everybody Specialisation (University of Michigan)
- Blockchain, Crypto assets, Decentralized Finance (INSEAD)
- Foundations of User Experience (UX) Design (Google)
- Global Financial Markets and Instruments (Rice University)
- Virtual experience programs: Lyft Back-end Engineering, Visa Token Service Technology

WEBSITES AND SOCIAL LINKS

Portfolio:
ayesharahman2002.github.io/portfolio/

LinkedIn:
linkedin.com/in/ayesha-rahman-ml/

GitHub:
github.com/AyeshaRahman2002

Neural Reconstruction of Heritage Artifacts
Built an end-to-end 3D reconstruction pipeline using NeRF, Nerfstudio and COLMAP, with multimodal vision-language interaction using BLIP/CLIP. Added agentic retrieval, semantic search, automated re-rendering, evaluation metrics, Docker deployment and interactive reporting.

SimMind: Cognitive Multi-Agent Simulation Framework
Designed a modular multi-agent framework with planner, executor, critic and coordinator agents in a deterministic CPU simulation environment. Implemented memory systems, FAISS vector retrieval, hallucination guards, task DSLs, reproducible seeded scenarios, FastAPI, Docker and Streamlit UI.

Multi-Modal CNN-RNN Framework for Visual Recognition and Captioning
Developed a CNN-RNN image recognition and captioning pipeline combining convolutional encoders with recurrent decoders. Improved TinyImageNet30 classification through augmentation, dropout, hyperparameter optimization, transfer learning and transformer-based semantic embeddings.

Edge-Optimized Robotic Vision & Navigation System
Optimized PyTorch ResNet18 for robotic edge vision and integrated LiDAR-camera fusion with ROS2 for obstacle avoidance and navigation. Implemented PID control, heuristic window prediction, LaserScan processing and deployment/testing across simulated and real robotic environments.

Secure Carbon Footprint Monitoring Platform
Designed a cloud-enabled platform for monitoring carbon emissions across smartphones, IoT devices and cloud usage. Implemented dashboards, authentication, HTTPS, encryption, session management, multi-device integration, multilingual support and privacy-preserving data handling.

Secure Full-Stack AI/ML Deployment Platform
Deployed TensorFlow models using Flask, React, AWS EC2, S3 and CloudFront for scalable model hosting. Built secure low-latency inference APIs with HTTPS, Flask-Bcrypt, CORS, load testing and end-to-end model-serving workflows.

Evolutionary Neural Optimization for Black-Box Learning
Implemented evolutionary algorithms including PSO, MPA and LM-IMPA to optimize neural network weights and architectures without gradient descent. Built benchmarking pipelines across synthetic optimization functions and real-world classification tasks, demonstrating strong performance on multimodal landscapes.

Interactive 3D Rendering Application
Built a C++/OpenGL 3D rendering application with custom matrix/vector maths, shader-based rendering, OBJ terrain loading, first-person camera control, orthophoto texturing, instancing, Blinn-Phong lighting and animated rocket launch simulation.

NLP Sentiment Analysis for Predictive Stock Market Signal
Developed an NLP sentiment analysis pipeline to classify customer reactions from social media and reviews. Correlated sentiment signals with discretized stock-price movements to explore predictive relationships between public sentiment and financial trends.